

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1190

Cajalco Tap 100 kV
September 29, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1190. The Interconnection Customer proposes a 224.4 MW solar plus storage facility interconnecting at the Cajalco Tap 100 kV substation in the Illinois area. The request was received on April 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 391 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,319,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SHEPHERDS NORTH / CAJALCO TAP 100 kV
Contingency: RIO LINDA / CAJALCO TAP 100 kV
Loading: 193.3% (Rating: 106 MVA)
DFAX: 100.0%
Pre-project loading: 19.3%
2. Facility: RIO LINDA SUBSTATION / CAJALCO TAP 100 kV Line
Contingency: SHEPHERDS NORTH / CAJALCO TAP 100 kV
Loading: 177.4% (Rating: 115 MVA)
DFAX: 100.0%
Pre-project loading: 17.7%
3. Facility: FLUSHING / RIO LINDA SUBSTATION 100 kV Line
Contingency: CAJALCO TAP / SHEPHERDS N 100 kV
Loading: 168.1% (Rating: 118 MVA)
DFAX: 100.0%
Pre-project loading: 22.0%
4. Facility: JOHNSON 1 / SHEPHERDS NORTH 100 kV
Contingency: CAJALCO TAP / RIO LINDA 100 kV
Loading: 160.3% (Rating: 122 MVA)
DFAX: 100.0%
Pre-project loading: 24.3%
5. Facility: CARLYSS / BOCA RATON 230 kV
Contingency: CARLYSS / TOP CROP SUBSTATION 500 kV line
Loading: 108.8% (Rating: 579 MVA)
DFAX: 33.5%
Pre-project loading: 95.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SHEPHERDS NORTH / CAJALCO TAP 100 kV	Rebuild line with higher-capacity conductor	\$3,149,000	100% (\$3,149,000)
RIO LINDA / CAJALCO TAP 100 kV	Rebuild line with higher-capacity conductor	\$5,915,000	100% (\$5,915,000)
FLUSHING / RIO LINDA SUBSTATION 100 kV Line	Rebuild line with higher-capacity conductor	\$3,754,000	100% (\$3,754,000)
JOHNSON 1 / SHEPHERDS NORTH 100 kV	Rebuild line with higher-capacity conductor	\$27,000,000	100% (\$27,000,000)
CARLYSS / BOCA RATON 230 kV	Reconductor line	\$15,102,000	96% (\$14,501,000)
	Total allocated to project		\$54,319,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.